

Trained Drone-Detection Models and Ground-Truth Test Datasets

Prepared June 1, 2026. Focus: detecting drones/UAVs as objects in sky/scene imagery - not aerial-view datasets like VisDrone that mainly detect cars/people from a drone camera.

How to use this file

Use the trained checkpoints/hosted models below for inference, then evaluate predictions against one of the listed ground-truth datasets. For fair mAP/precision/recall testing, normalize class names first: for example, map UAV, drone, Drone, and Fix-wing-drone to the class convention used by your evaluation script.

1) Published trained models/checkpoints

Trained model / access	Base and classes	Checkpoint status	Ground-truth dataset(s) to test without training	Practical notes
doguilmak/Drone-Detection-YOLOv8x Hugging Face model best.pt	YOLOv8x; single class: drone.	Downloadable PyTorch/Ultralytics weight: weight/best.pt .	- Kaggle YOLO Drone Detection Dataset - Det-Fly images + annotations - DUT Anti-UAV img + gt - Roboflow YOLOv8_DetFly dataset/model	Good first baseline for single-class drone detection. Expect performance loss on tiny long-range targets unless image resolution/tiling is tuned.
doguilmak/Drone-Detection-YOLOv11x Hugging Face model	YOLOv11x; single class: drone.	Published trained model on Hugging Face. Model card states it was trained for drone detection.	- Kaggle YOLO Drone Detection Dataset - DUT Anti-UAV - Anti-UAV300/410/600	Larger model; better accuracy potential, heavier GPU cost. Use this if real-time CPU/edge inference is not the first constraint.
doguilmak/Drone-Detection-YOLOv7 Hugging Face model best.pt	YOLOv7; single class: drone.	Downloadable weight: weight/best.pt .	- Kaggle YOLO Drone Detection Dataset - Det-Fly - Roboflow DroneDet	Useful older baseline. For a new pipeline, YOLOv8/YOLOv11 is usually easier to maintain with current Ultralytics tools.
Alireza0K/Unmanned-Aerial-Vehicle GitHub repo	YOLOv8; classes: drone and Fix-wing-drone.	Repo states it includes trained model weights; listed weight name: 51ep-16-GPU.pt .	- Repo dataset includes train/valid/test split: 1000 train, 200 val, 300 test images. - Det-Fly - AOD-4 airborne object dataset	Best option here if you need both multirotor and fixed-wing UAV categories. When comparing to single-class datasets, merge both outputs into drone/UAV.
Roboflow YOLOv8_DetFly(02) Hosted model + dataset	Hosted object-detection model; class: UAV.	Roboflow-hosted trained model/API, not a simple local .pt download from the public page.	- Same Roboflow project includes 6,913 labeled UAV images. - Original Det-Fly repo - DUT Anti-UAV	Quickest no-training option if hosted inference/API is acceptable. Page reports mAP@50 = 92.7%, precision = 97.4%, recall = 88.8% on its project split.
Roboflow DroneDet Hosted model + dataset	Hosted object-detection model; class: drone.	Roboflow-hosted trained model/API.	- Same project includes 1,339 open-source drone images. - Kaggle YOLO Drone Detection Dataset - Det-Fly	Good lightweight hosted baseline. Page reports mAP@50 = 95.4%, precision = 96.1%, recall = 88.6% on its project split.
Javvanny/yolov8m_flying_objects_detection Hugging Face model best.pt	YOLOv8m; classes: Drone, Airplane, Helicopter, Bird, Background.	Downloadable best.pt in yolov8m/weights.	- AOD-4: airplane, helicopter, drone, bird - Drone-detection-dataset: video labels/classes - WOSDETC Drone-vs-Bird annotations/images	Most useful when birds, aircraft, or helicopters create false positives. For pure anti-drone evaluation, report drone AP separately.

2) Ground-truth datasets useful for no-training evaluation

The datasets below have ground-truth labels/annotations or project splits you can use to test the trained models above. Some are image datasets, some are video benchmarks; for video datasets, evaluate frame-by-frame after extracting frames or use the benchmark's native evaluation code.

Dataset	Ground truth / labels	Why it is useful	Best matching model(s)
Kaggle YOLO Drone Detection Dataset Dataset link	Bounding boxes for drone detection; used/linked by the doguilmak YOLOv7/YOLOv8x model cards.	Direct single-class drone evaluation with YOLO-style annotations.	doguilmak YOLOv8x, YOLOv7, YOLOv11x; Roboflow DroneDet.
Det-Fly GitHub repo	Repo provides separate downloads for JPEGImages and Annotations.	Air-to-air micro-UAV detection; strong match for detect-another-drone scenarios.	YOLOv8x/YOLOv7/YOLOv11x; Roboflow YOLOv8_DetFly.
DUT Anti-UAV GitHub repo	GitHub provides image and ground-truth downloads: img + gt.	Drone detection/tracking benchmark; useful for small and difficult UAV targets.	Single-class drone models; map labels to drone/UAV.
Anti-UAV300/410/600 Official repo	Full-HD RGB/IR video sequences with bounding boxes, attributes, and target-exists flags.	Good for detection plus tracking/failsafe experiments; includes absent-target frames.	YOLO detector plus tracker; use RGB subset for visible-camera models.
Roboflow YOLOv8_DetFly(02) Dataset/model	6,913 labeled UAV images; class UAV; CC BY 4.0 project page.	Easy download/export to YOLO formats and hosted model comparison.	Roboflow hosted DetFly model, YOLOv8x/YOLOv11x after class-name mapping.
Roboflow DroneDet Dataset/model	1,339 open-source drone images; class drone; CC BY 4.0 project page.	Small, convenient sanity-test dataset for drone-only detectors.	Roboflow DroneDet; doguilmak models.
AOD-4 airborne object dataset Mendeley Data	22,516 images across airplanes, helicopters, drones, and birds.	Best for measuring false positives between drones, birds, helicopters, and airplanes.	Javvanny YOLOv8m flying-objects model; Alireza0K if evaluating UAV categories only.
Drone-detection-dataset GitHub repo	650 visible/IR videos plus label files; classes include Airplane, Bird, Drone, Helicopter. Python label reader is documented.	Good for video and sensor-fusion experiments; also has absent-target cases.	Javvanny YOLOv8m; detector plus temporal tracker.
WOSDETC Drone-vs-Bird Challenge GitHub annotations/images	Official repo contains annotations/images; challenge focuses on small drones versus birds.	Very relevant if your detector triggers on birds. Some challenge datasets may require following the challenge site's data-use instructions.	Javvanny YOLOv8m; any single-class drone detector for false-alarm testing.
VisioDECT ScienceDirect article	20,924 annotated multi-drone images; annotations in TXT, XML, and CSV are described by the article.	Modern multi-drone detection benchmark with multiple real-world conditions.	Single-class drone/UAV detectors; useful for multi-instance drone scenes.

3) Recommended test order

For your SKYSCouter/drone project, test in this order: (1) start with the Alireza0K YOLOv8 model and doguilmak YOLOv8x/YOLOv11x on Det-Fly and DUT Anti-UAV; (2) add AOD-4 or WOSDETC Drone-vs-Bird to measure false alarms on birds/aircraft; (3) if hosted inference is acceptable, compare against Roboflow YOLOv8_DetFly(02) and DroneDet as fast baselines; (4) use your own camera footage as an external holdout set before choosing a model for onboard deployment.

Notes and limitations

Some repositories publish weights but do not clearly publish the exact training split. Treat their reported metrics as project-level indicators, not as proof of performance on your camera. Also, Roboflow hosted models may not expose a public local .pt checkpoint from the web page; they are still trained models that can be tested through the hosted API or workspace export options where available.

Primary source links checked

- [doguilmak YOLOv8x model card and best.pt](#)
- [doguilmak YOLOv7 model card and best.pt](#)
- [doguilmak YOLOv11x model card](#)
- [Alireza0K Unmanned-Aerial-Vehicle GitHub repo](#)
- [Javvanny YOLOv8m flying-objects model](#)
- [Roboflow YOLOv8_DetFly\(02\)](#)
- [Roboflow DroneDet](#)
- [Det-Fly GitHub](#)
- [DUT Anti-UAV GitHub](#)
- [Anti-UAV official repo](#)
- [AOD-4 Mendeley Data](#)
- [Drone-detection-dataset GitHub](#)
- [WOSDETC Drone-vs-Bird challenge](#)